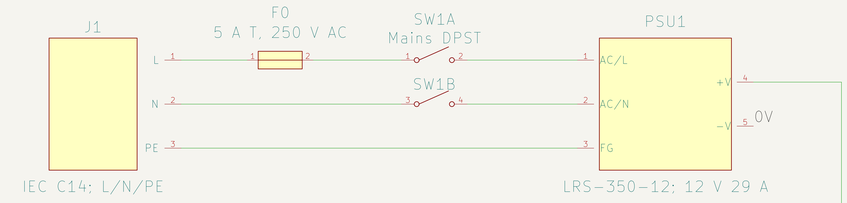
KIRO SO-101 MOBILE DUAL-ARM ROBOT | ELECTRICAL SCHEMATIC (rev E)

1 AC INPUT (mains - unplug before work)



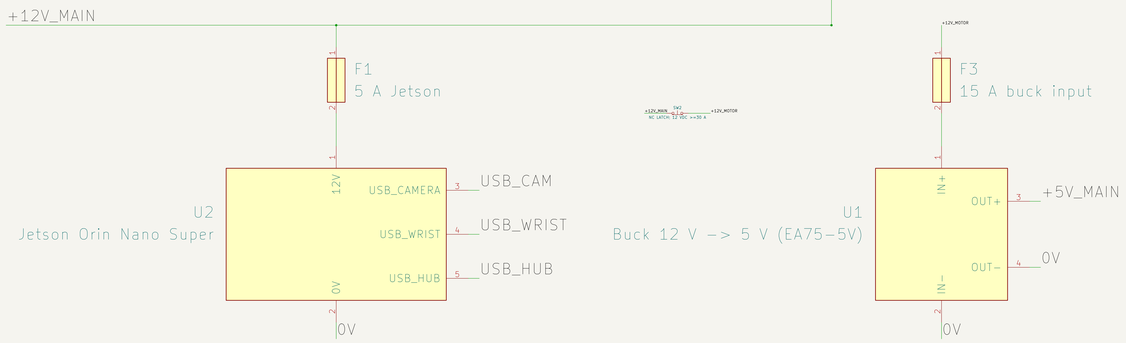
POWER: 12 V (+12V_MAIN) feeds Jetson Orin Nano and the four wheel servos (ST3215-C018).

5 V (+5V_MAIN, from U1 buck) feeds both arms: leader + follower servos (STS3215).

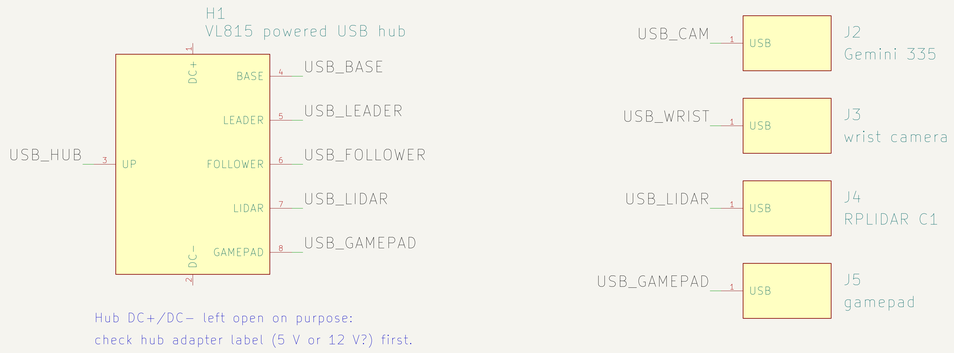
SW2 MOTOR STOP (prototype): latching NC, 12 V DC motor-load break rating 30 A or more.

Opens F2 wheels and F3 buck input; F1 Jetson stays on. Not safety-rated.

2 12 V DISTRIBUTION + COMPUTE + ARM BUCK

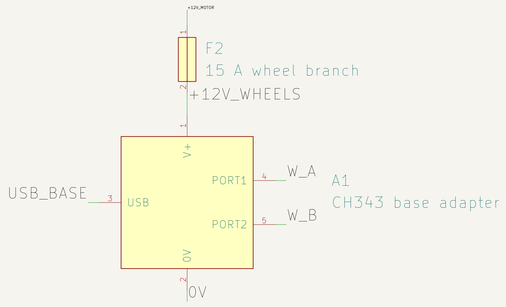


3 USB LOGIC LINKS

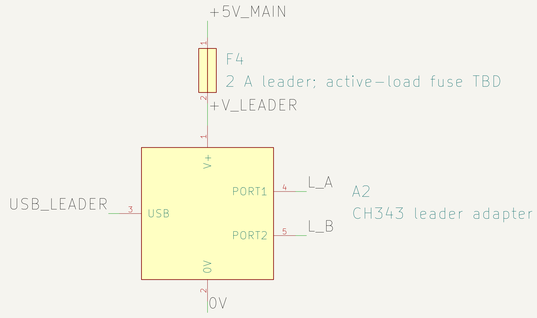


BOTH ARMS ACTIVE: installed EA75-5V buck is 5 V / 15 A TOTAL. Verify combined arm stall current; upgrade buck, branch fuses and wiring before loaded two-arm operation.

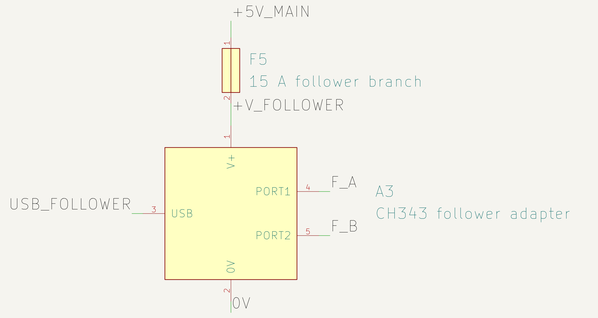
4 WHEELS 12 V (ST3215-C018)



5 LEADER ARM +5V_MAIN (STS3215)



6 FOLLOWER ARM +5V_MAIN (STS3215-C001)



Kiro SO-101 project

Sheet: /

File: KIRO_FINAL_Kicad.sch

Title: Kiro robot - complete two-arm electrical schematic

Size: A2 | Date: 2026-10-27 | **Rev: E**

KiCad 8.0.4 10.0 | 1/1